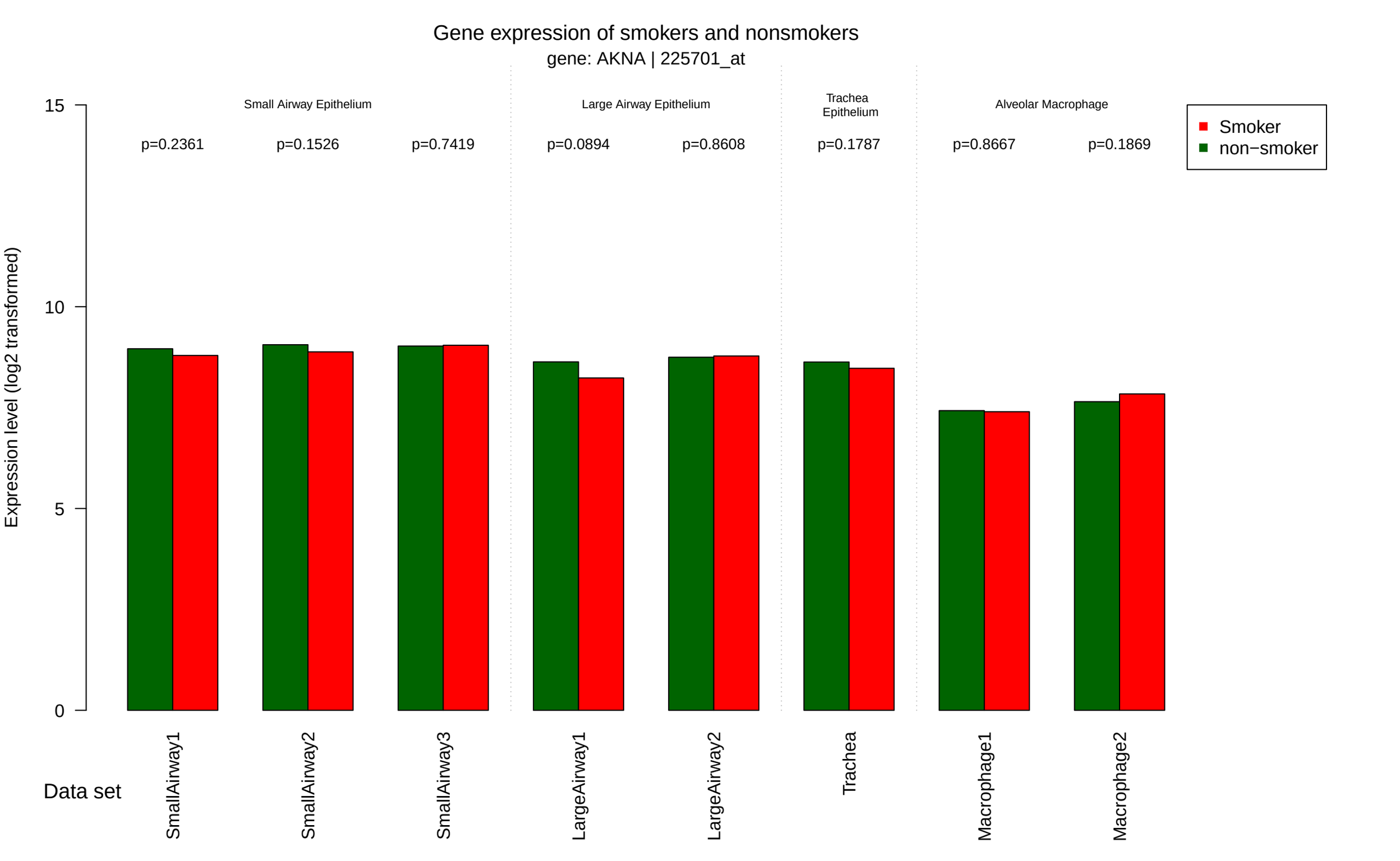
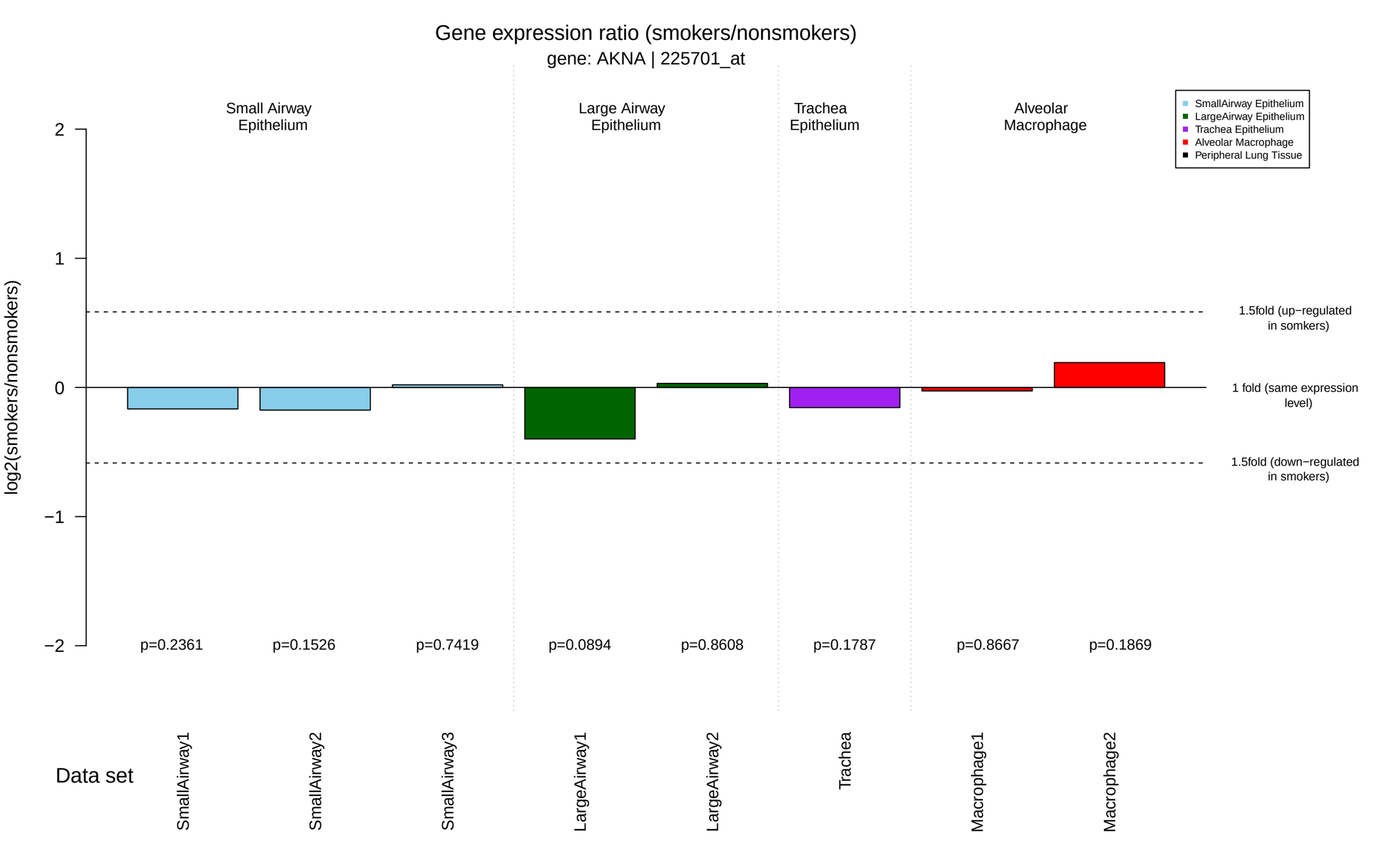
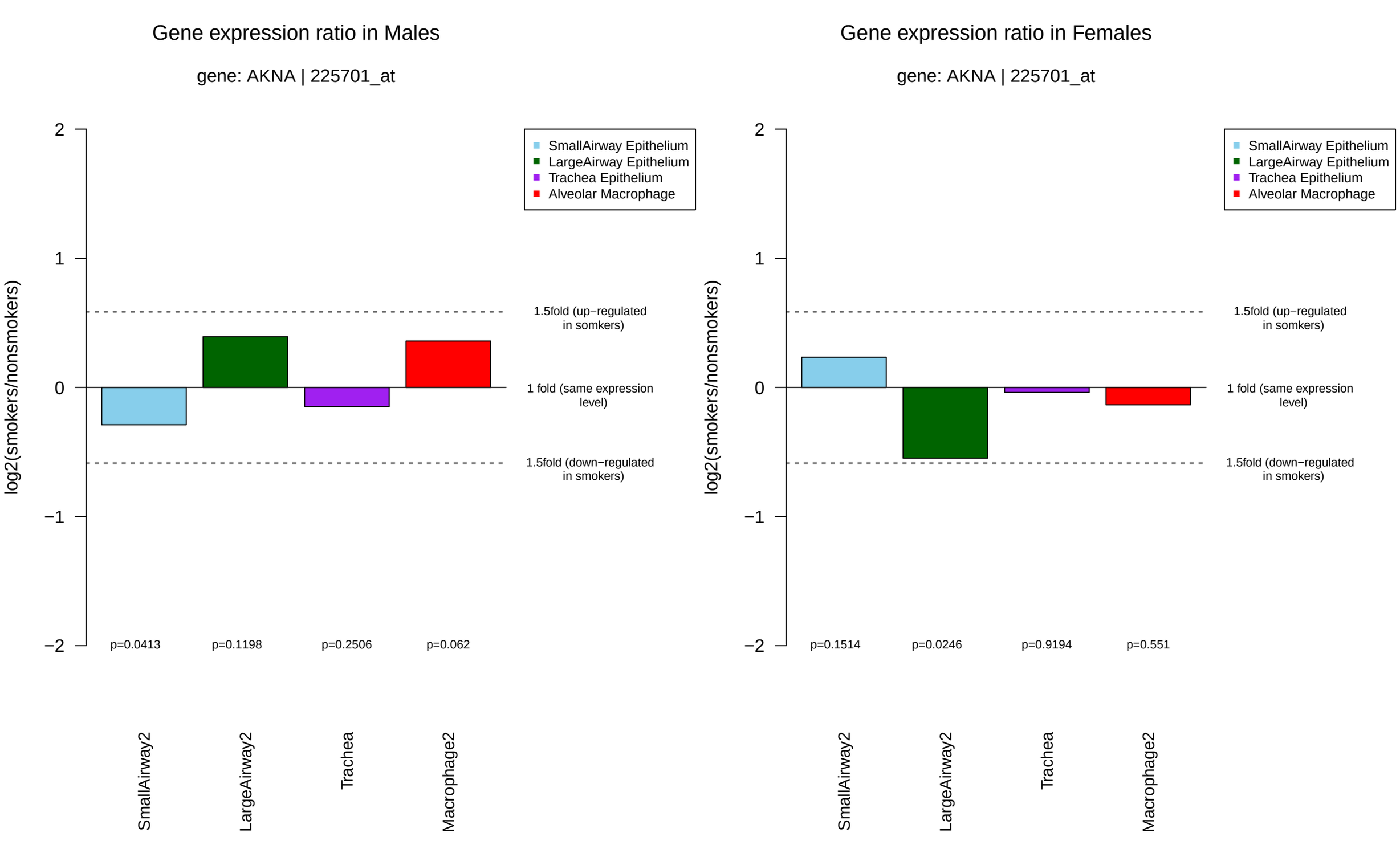
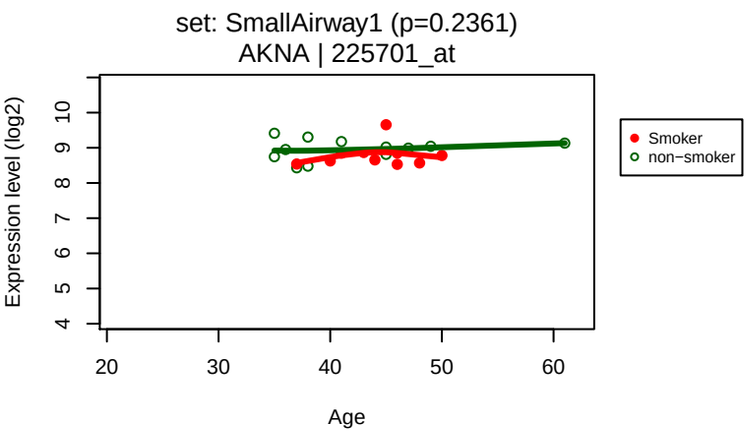
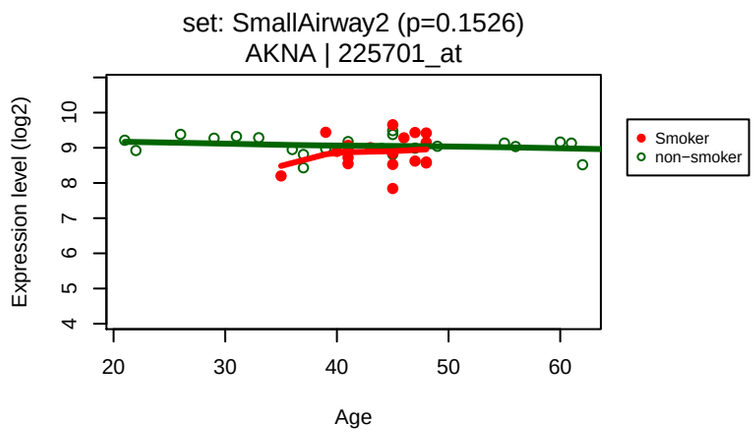
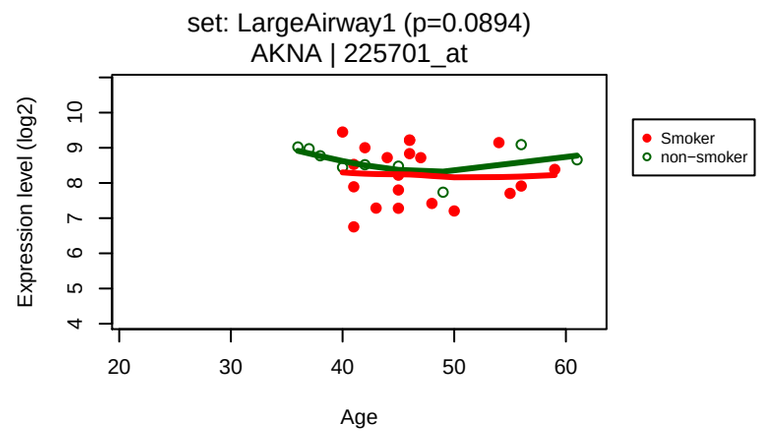

Gene expression of smokers and nonsmokers
gene: AKNA | 225701_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2361
p=0.1526
p=0.7419
p=0.0894
p=0.8608
p=0.1787
p=0.8667
p=0.1869
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

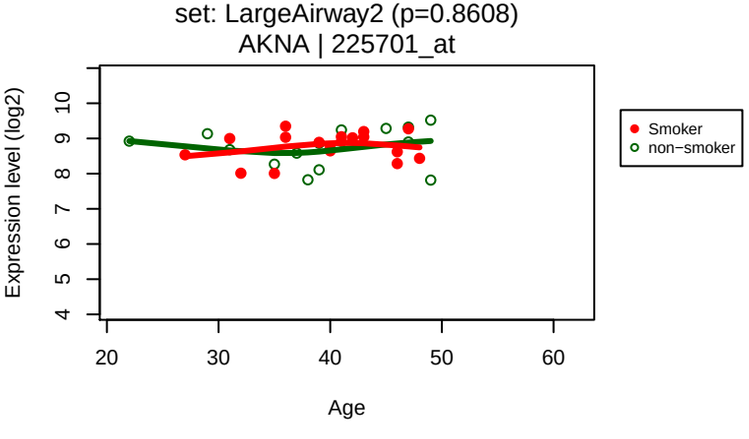

Gene expression ratio (smokers/nonsmokers)
gene: AKNA | 225701_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2361
p=0.1526
p=0.7419
p=0.0894
p=0.8608
p=0.1787
p=0.8667
p=0.1869
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

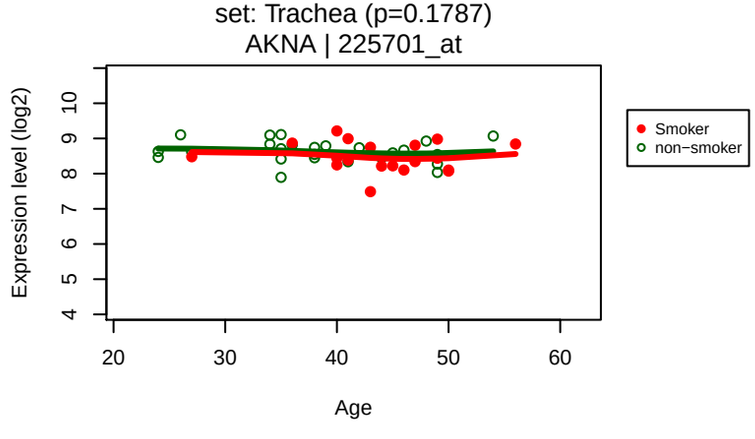

Gene expression ratio in Males
gene: AKNA | 225701_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0413
p=0.1198
p=0.2506
p=0.062
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKNA | 225701_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1514
p=0.0246
p=0.9194
p=0.551
SmallAirway2
LargeAirway2
Trachea
Macrophage2

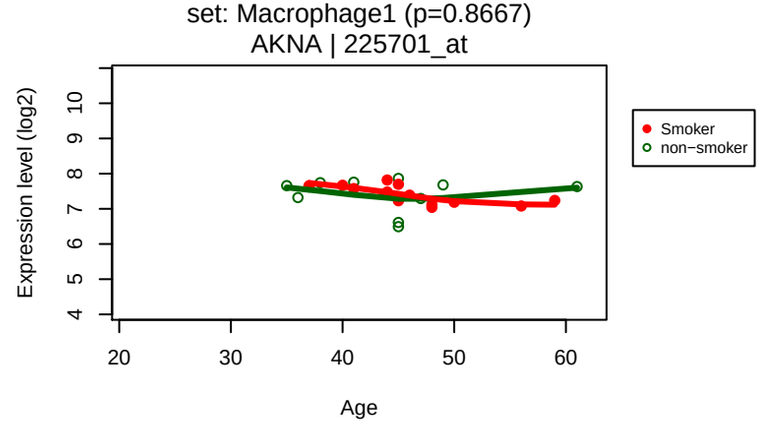

set: SmallAirway1 (p=0.2361)
AKNA | 225701_at
Expression level (log2)
Age
Smoker
non−smoker

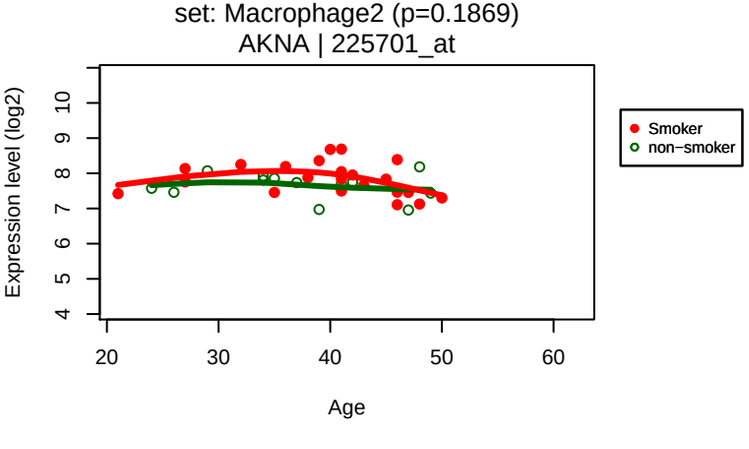

set: SmallAirway2 (p=0.1526)
AKNA | 225701_at
Expression level (log2)
Age
Smoker
non−smoker

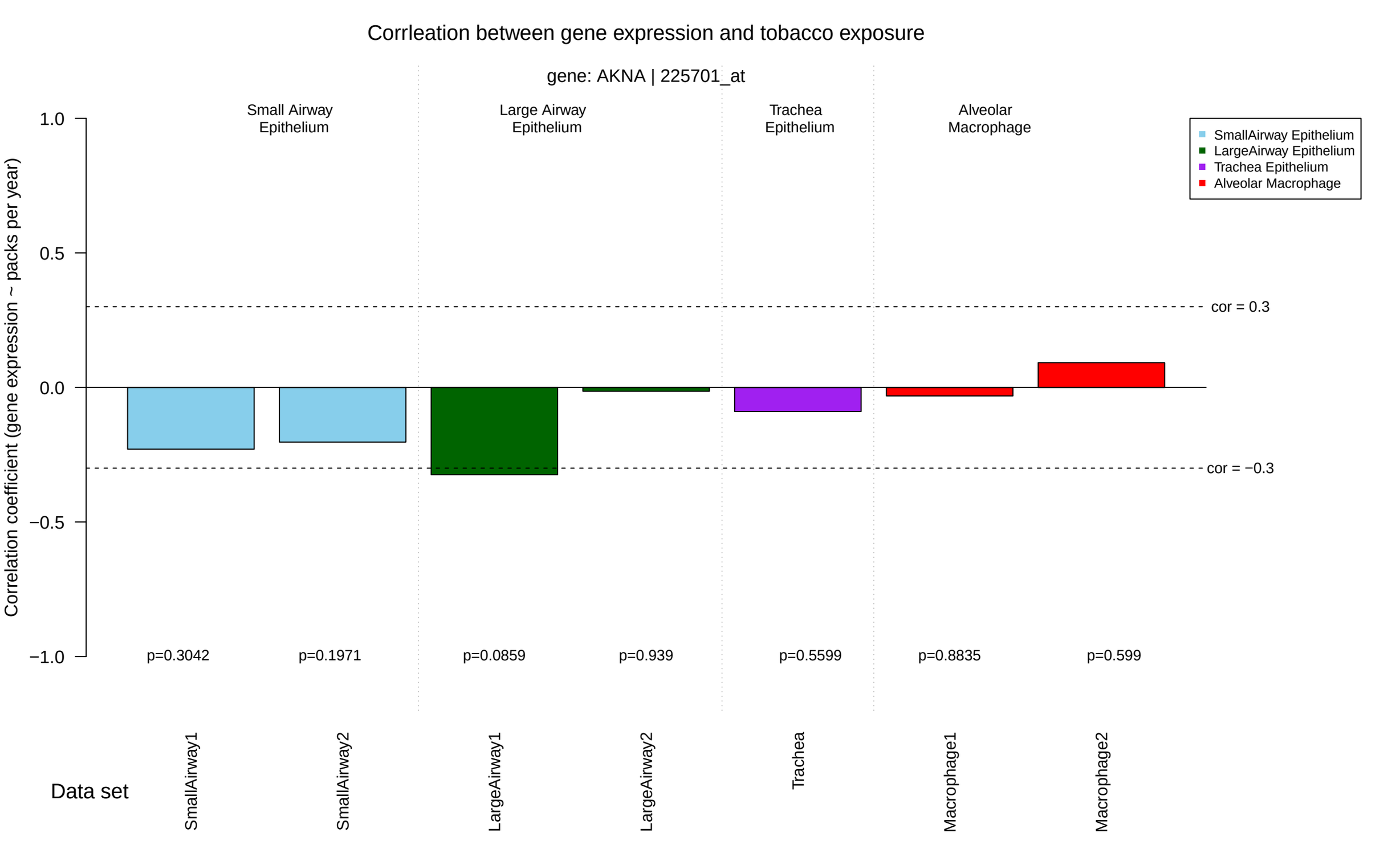

set: LargeAirway1 (p=0.0894)
AKNA | 225701_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8608)
AKNA | 225701_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1787)
AKNA | 225701_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8667)
AKNA | 225701_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1869)
AKNA | 225701_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKNA | 225701_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3042
p=0.1971
p=0.0859
p=0.939
p=0.5599
p=0.8835
p=0.599
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2